

EXPERIMENT 16: PROLOG PROGRAM TO SUM INTEGERS FROM 1 TO N

Aim

To write a Prolog program to find the sum of integers from 1 to N.

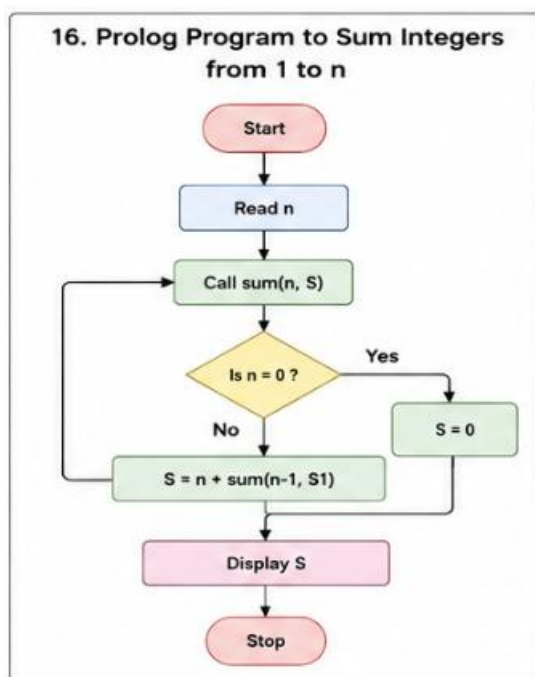
Objective

To calculate the sum of numbers from 1 to a given number using recursion in Prolog.

Algorithm

1. Read N.
2. If N is 0, return 0.
3. Otherwise, add N to the sum of N-1.
4. Display the sum.

Flowchart



CODE

```
n = int(input("Enter n: "))
```

```
sum = 0
```

```
for i in range(1, n + 1):
```

```
sum = sum + i
```

```
print("Sum =", sum)
```

OUTPUT

```
Enter n: 25  
Sum = 325
```

Result

The sum of integers from 1 to N was calculated successfully.

Conclusion

The program demonstrates recursion in Prolog to calculate the sum of integers.